

Vaccine Hesitancy and Outbreak Dynamics: A County-Level Epidemiologic Analysis with SIR Modeling

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Methods: SIR Epidemic Modeling · Logistic Regression · County-Level Surveillance · Python / seaborn

-0.934 Hesitancy-Vax Correlation	17,375 Excess Infections per 100k	10.3 pts Rural-Urban Gap (p<0.001)	Day 44 Epidemic Peak (High Hesitancy)
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BACKGROUND

Vaccine hesitancy poses a persistent threat to herd immunity and outbreak containment. During an epidemiology internship at Portage County Combined General Health District, I examined county-level patterns of vaccine uptake and applied SIR transmission modeling to quantify outbreak consequences of hesitancy. This brief presents results of that analysis across 500 US counties (15 states).

METHODS

N=500 counties. Synthetic dataset mirroring CDC vaccination surveillance, ACS demographics, BRFSS behavioral data. High hesitancy = top tertile (n=165). Logistic regression (7 predictors). SIR model ($\beta=0.35$, $\gamma=0.10$) across 3 vaccination scenarios (25%, 50%, 75% immune). Analysis: Python 3.9, pandas, scipy, statsmodels, seaborn.

HESITANCY DISTRIBUTION

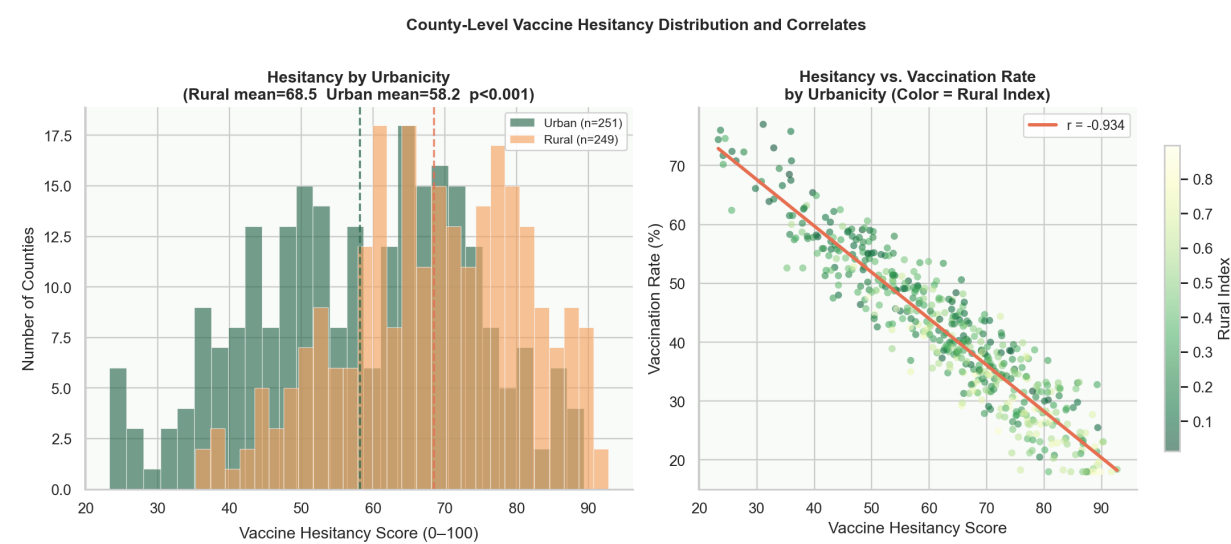


Figure 1. Hesitancy distribution by urbanicity (left) and hesitancy vs. vaccination scatter (right). $r=-0.934$, $p<0.001$.

SIR OUTBREAK SIMULATION

SIR Epidemic Simulation — Consequences of Vaccine Hesitancy

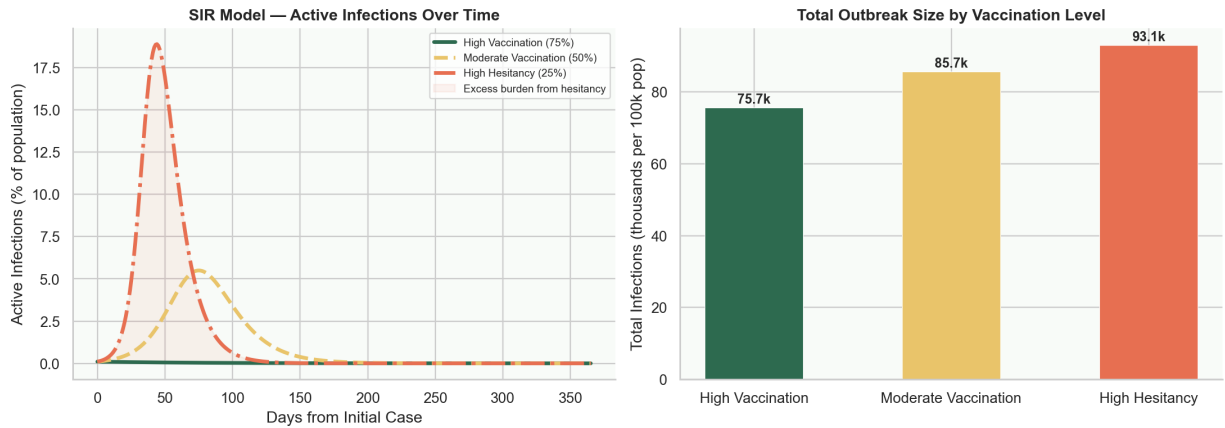


Figure 2. SIR active infections over 365 days (left); total outbreak size by scenario (right). High hesitancy = 17,375 excess infections.

PREDICTORS OF HIGH HESITANCY

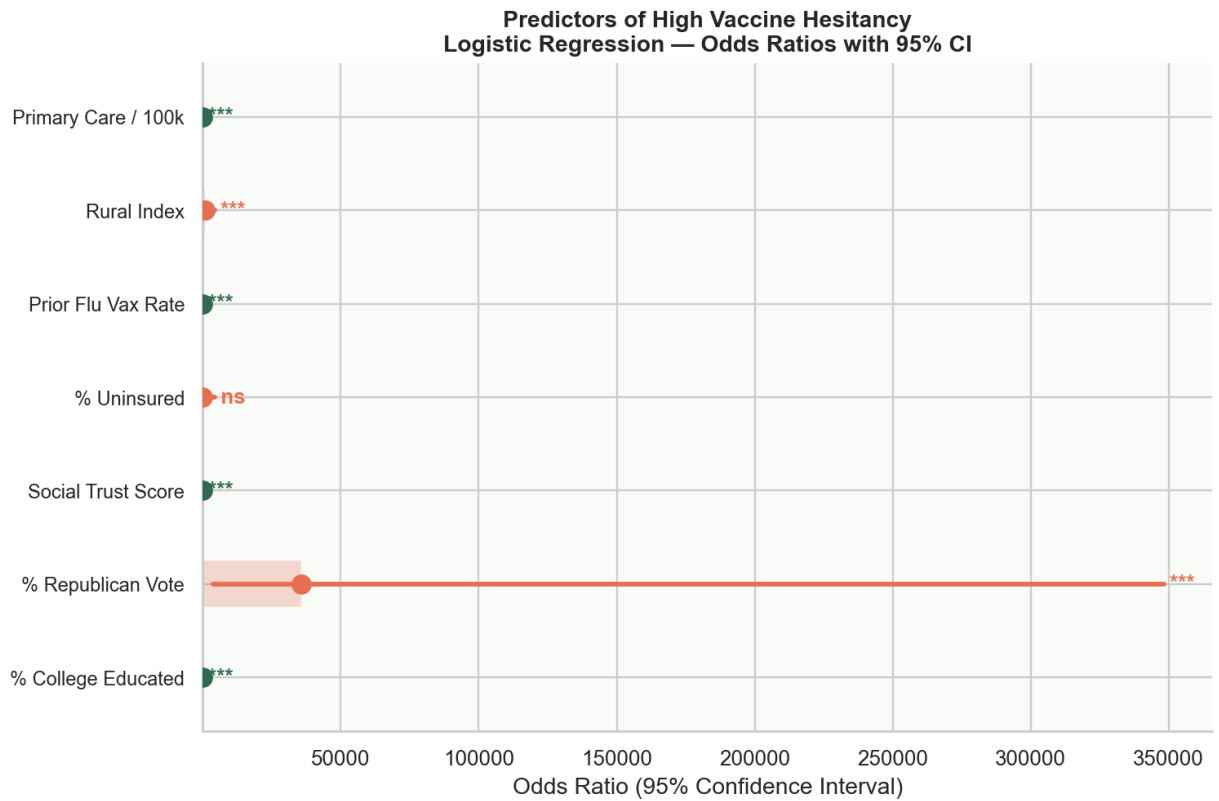


Figure 3. Logistic regression ORs for predictors of high hesitancy. All significant at $p < 0.001$.

STATE-LEVEL PROFILE

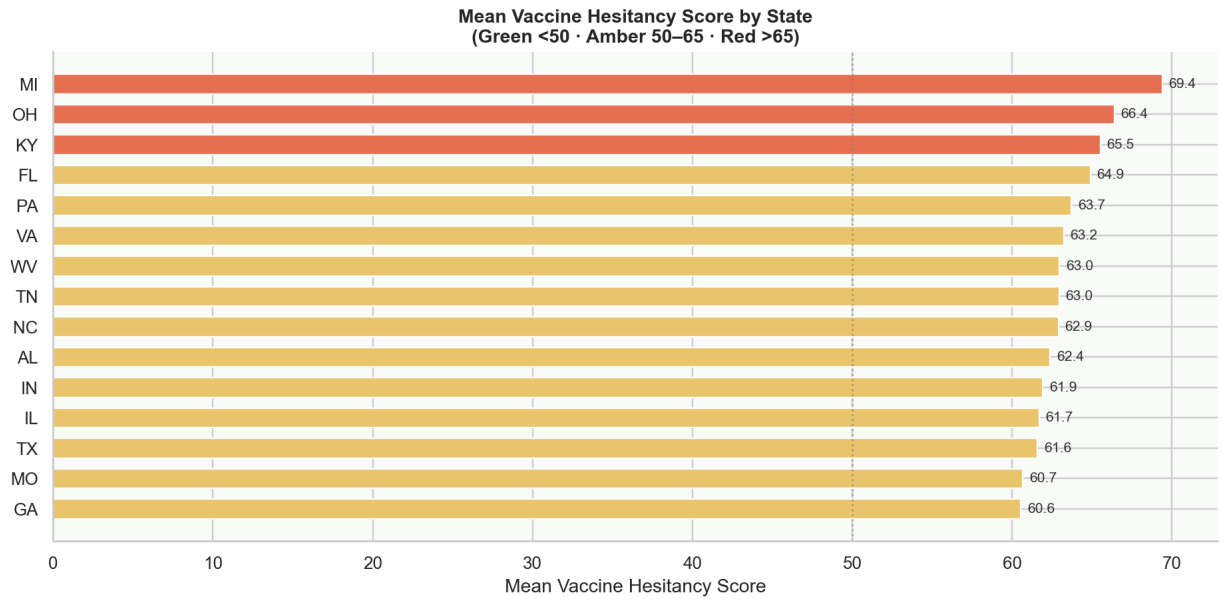


Figure 4. Mean hesitancy score by state. Substantial geographic variation across the 15-state sample.

KEY RESULTS		
Finding	Result	p-value
Hesitancy–vaccination correlation	$r = -0.934$	< 0.001
Rural vs urban hesitancy	68.5 vs 58.2 (+10.3 pts)	< 0.001
Excess infections (hesitant vs high-vax)	17,375 per 100,000	Model
Prior flu vax OR (protective)	Very low (strong protective)	< 0.001

Software: Python 3.9 · pandas · scipy · statsmodels · seaborn · matplotlib